

RAMBØLL	PROJECT —	PROJECT NO. —	PREPARED —
	Rectangular Beam Bending Check		CHECKED —
REV / DATE 0.1 · 2026-07-20			APPROVED —
			DRAFT

PURPOSE

Elastic bending stress verification for a simply supported rectangular timber beam under uniform load.

1 Input Parameters

Symbol	Value	Unit	Description
b	200	mm	Section width
h	400	mm	Section depth
L	5000	mm	Span
q	4.5	kN/m	Distributed load (design)
f_m	24	MPa	Design bending strength

2 Calculation

Maximum bending moment

$$\begin{aligned} M_{max} &= \frac{q \cdot (\frac{L}{1000})^2}{8} \\ &= \frac{4.5 \cdot (\frac{5000}{1000})^2}{8} \\ &= \mathbf{14.06 \text{ kNm}} \end{aligned}$$

Elastic section modulus

$$\begin{aligned} W &= \frac{b \cdot h^2}{6} \\ &= \frac{200 \cdot 400^2}{6} \\ &= \mathbf{5.333e + 06 \text{ mm}^3} \end{aligned}$$

Maximum bending stress

$$\begin{aligned} \sigma_m &= M_{max} \cdot 1000000 \cdot \frac{1}{W} \\ &= 14.06 \cdot 1000000 \cdot \frac{1}{5.333e + 06} \\ &= \mathbf{2.637 \text{ MPa}} \end{aligned}$$

3 Verification

OK

$\sigma_m = 2.637 \text{ MPa} \leq f_m = 24 \text{ MPa}$